DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023-24
MONDAY TEST

CLASS: X | **FULL MARKS: 40**
SUBJECT: PHYSICS | **DATE: 20.11.2023**

General Instructions:

- **This paper consists of <u>three</u> printed pages.**
- **All questions are <u>compulsory</u>.**
- **Marks will be deducted for spelling errors.**

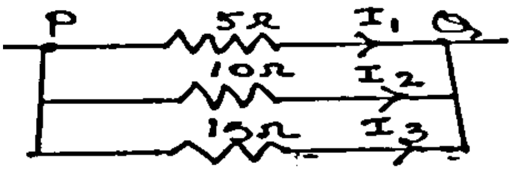
SECTION A

(Attempt all questions.)

Question 1

Choose the correct answers to the questions from the given options. **[1 × 7 = 7]**

(i) With reference to the diagram shown below, find the correct option.



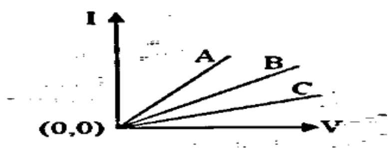
a) $I_1 = 2I_2 = 3I_3$ **b)** $I_1 = 4I_2 = 3I_3$ **c)** $2I_1 = I_2 = 3I_3$ **d)** $3I_1 = 2I_2 = I_3$

(ii) If the current flowing through a fixed resistor is halved, the heat produced in it will become
a) **double** **b) half** **c) one-fourth** **d) four times**

(iii) **A copper wire of resistance R is bent to form a circular coil. The resistance across any two diametrically opposite points will be**
a) R/4 **b) R/2** **c) R** **d) 2R**

(iv) Assertion (A): Tungsten is used for making filaments of incandescent lamps.
Reason (R): The melting point of tungsten is very low.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

(v) The graph shows the variation of current I against V relation for three conductors A, B and C. Choose the correct relation for the resistors A, B and C.



a) $R_A > R_B > R_C$ **b)** $R_B > R_C < R_A$ **c)** $R_C > R_B < R_A$ **d)** $R_C > R_B > R_A$

(vi) When number of resistors are connected in parallel, the resultant resistance of the combination is ____.
a) Average of all the resistances **b) Greater than the greatest**
c) Less than the least **d) The sum of the least and the greatest**

(vii)The rise in temperature of fuse depends on

a) current **b) radius** **c) length of wire** **d) Both (a) & (b)**

Question 2

(i) a) Why is a double pole switch used as the main switch in distribution board?

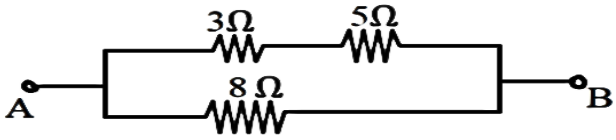
b) Why is household electric connection done in parallel?

c) Mention any one factor that affects the internal resistance of a cell. **[3]**

(ii) An electric bulb is rated 220 V and 100 W. What will be its power consumption when it is operated at 110 V? **[2]**

(iii) An electric kettle is rated 3 kW - 250 V. Justify whether this kettle can be used in a circuit that contains a fuse of current rating 13 A. **[2]**

(iv) Calculate the resistance across AB in the circuit given below. **[2]**



(v) How does an increase in the temperature affect the conductance of a

(a) metal (b) semiconductor? **[2]**

(vi) How can you connect three resistors 2 Ω, 3 Ω and 5 Ω so that the same current flows through each of them, given that you can't join all three of them in series? Draw the required arrangement of resistors. **[2]**

SECTION B

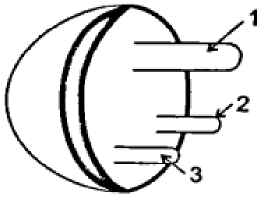
(Attempt all questions.)

Question 3 **[3 + 3 + 4 = 10]**

i) There are three pins in an electric plug top as shown in the figure below. Answer the following:

a) Why is the pin labelled 1 is longer and thicker?

b) In which of the three pins should the wire corresponding to electric switch be connected?



ii) Sonia has an electric kettle that takes 1 minute to boil four cups of water when connected to a 220 V power supply. She decides to visit Digha during her vacations. In Digha, she uses the same kettle to boil the same quantity of water but finds that the time taken there is 4 minutes.

(a) Assuming the room temperature to be the same at both the places, what reason can be attributed to the longer time taken in Digha? Justify your answer with proper mathematical calculation.

(b) Name one material that is used for the heating element of the kettle.

iii) (a) A metallic wire of resistance 30 Ω is stretched until its length is tripled. What will be the new resistance and resistivity? Appropriate mathematical calculation is to be shown.

(b) Name a substance that doesn't undergo any significant change in its specific resistance with temperature.

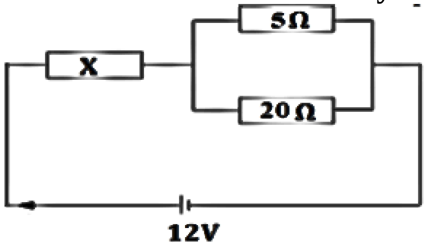
Question 4 **[3 + 3 + 4 = 10]**

i) An electric bulb is rated '240 V-100 W'.

(a) What information can you gather from the above statement?

(b) Find the energy consumed by the bulb in 10 minutes.

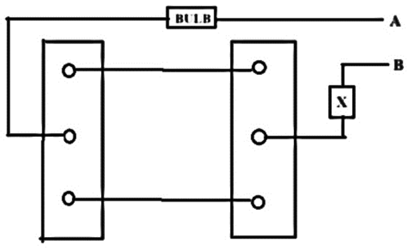
ii) Three resistors are connected to a 12V battery as shown in the figure given below:



The current through X is 1.2 A. Calculate the

(a) total resistance.
(b) magnitude of X
(c) current through 5Ω resistance

iii) The diagram shows a dual control switch connected to a bulb.



a) Copy the diagram and complete it so that the bulb is switched ON.
b) Identify the wires A and B. What should be the color code of wire A?
c) Identify X.